

MAT 1272 – Statistics

Measures of Dispersion for Ungrouped Data

Section 3.2

Range • Variance • Standard Deviation

TI-84 Standard Deviation

Based on the course textbook:
Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley

Section 3.2: Why Measure Dispersion?

Company 1

47, 38, 35, 40, 36, 45, 39

Mean = 40

Company 2

70, 33, 18, 52, 27

Mean = 40

Same center, different spread

Both data sets have the same mean, but Company 2 has much greater variation.

Dispersion, spread, and variation describe how observations are spread out.

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.

3.2.1 Range

Finding the range for ungrouped data

$$\text{Range} = \text{Largest value} - \text{Smallest value}$$

Example 3-13: Total Areas of Four States

53,182, 49,651, 69,903, 267,277

$$\text{Range} = 267,277 - 49,651 = 217,626 \text{ square miles}$$

Limitation

The range uses only the largest and smallest values and is sensitive to outliers.

3.2.2 Variance and Standard Deviation

Standard deviation

The standard deviation tells us how closely values are clustered around the mean.

Small standard deviation

Values are relatively close to the mean.

Large standard deviation

Values are spread over a larger range around the mean.

$$\text{Standard deviation} = \sqrt{\text{variance}}$$

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.

3.2.2 Population vs. Sample Notation

	Population	Sample
Variance	σ^2	s^2
Standard deviation	σ	s
Number of values	N	n
Mean	μ	$\bar{x}$

Important

Decide whether the data represent a population or a sample before choosing a formula or interpreting TI-84 output.

3.2.2 Basic Formulas

Population:

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Sample:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Deviation from the mean

$x - \mu$ or $x - \bar{x}$ measures how far an observation lies from the mean.

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.

3.2.2 Why Square the Deviations?

For the sample scores

$$82, 95, 67, 92, \quad \bar{x} = 84.$$

x	$x - \bar{x}$
82	-2
95	11
67	-17
92	8
$\sum (x - \bar{x}) = 0$	

Key point

Positive and negative deviations cancel. Squaring the deviations prevents this cancellation.

3.2.2 Short-Cut Formulas

Population variance:

$$\sigma^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{N}}{N}$$

Sample variance:

$$s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n - 1}$$

Then:

$$\sigma = \sqrt{\sigma^2}$$

$$s = \sqrt{s^2}$$

Do not confuse these two quantities

$$\sum x^2 \neq (\sum x)^2$$

Example 3-14: Sample Standard Deviation

Problem

The 2014 compensations (in millions of dollars) of 11 female CEOs of American companies are:

19.3, 16.2, 19.6, 19.3, 33.7, 21.0, 22.5, 16.9, 28.7, 42.1, 22.2.

Find the variance and standard deviation for these data.

Because these 11 CEOs are treated as a **sample**,

$$n = 11, \quad \sum x = 261.5, \quad \sum x^2 = 6849.07.$$

$$s^2 = \frac{6849.07 - \frac{(261.5)^2}{11}}{10} = \boxed{63.2502}, \quad s = \sqrt{63.2502} \approx \boxed{7.95}.$$

The sample standard deviation is about \$7.95 million.

Example 3-15: Population Standard Deviation

Problem

The following data give the 2015 earnings (in thousands of dollars) before taxes for **all six employees of a small company**:

88.50, 108.40, 65.50, 52.50, 79.80, 54.60.

Calculate the variance and standard deviation for these data.

all six employees $\implies$ population $\implies N = 6$.

The textbook obtains

$$\sigma^2 = \boxed{388.90}, \quad \sigma = \sqrt{388.90} = \boxed{19.721 \text{ thousand dollars}}.$$

Why σ ?

These observations are the entire population of employees, so we use the population standard deviation.

3.2.2 Important Properties

- 1 Variance and standard deviation are never negative.
- 2 If all observations are equal, both are zero.
- 3 Variance is measured in squared units.
- 4 Standard deviation has the same units as the original data.
- 5 Variance and standard deviation are nonresistant; outliers can increase them.

No variation

35, 35, 35, 35 $\implies$ variance = 0, standard deviation = 0.

TI-84: Enter the Data

Step 1

- 1 Press **STAT**.
- 2 Select **1:Edit**.
- 3 Enter the observations in **L1**.
- 4 Press **ENTER** after each value.

Practice data

Enter

82, 95, 67, 92

into **L1**.

Before calculating

Decide whether the data represent a sample or a population.

TI-84: Run 1-Var Stats

Step 2

- 1 Press **STAT**.
- 2 Arrow right to **CALC**.
- 3 Select **1:1-Var Stats**.
- 4 Enter **L1** if needed.
- 5 Select **Calculate**; press **ENTER**.

The output includes

$$\bar{x}, \quad S_x, \quad \sigma_x, \quad n.$$

Read the correct standard deviation

$$S_x = \text{sample SD}$$

$$\sigma_x = \text{population SD}$$

TI-84: Which Standard Deviation?

Sample data	Population data
Use S_x	Use σ_x
Corresponds to s	Corresponds to σ
Variance uses $n - 1$	Variance uses N

Important

Choose S_x or σ_x according to what the data represent.

TI-84 Practice

Enter

88.50, 108.40, 65.50, 52.50, 79.80, 54.60

in $L1$.

- 1 Run **1-Var Stats**.
- 2 Find $\bar{x}$, S_x , and σ_x .
- 3 These are all six employees. Which standard deviation should we report?

Check

For this population, report

$$\sigma_x \approx 19.721 .$$

Section 3.4.2: Empirical Rule

For a **bell-shaped distribution**, approximately:

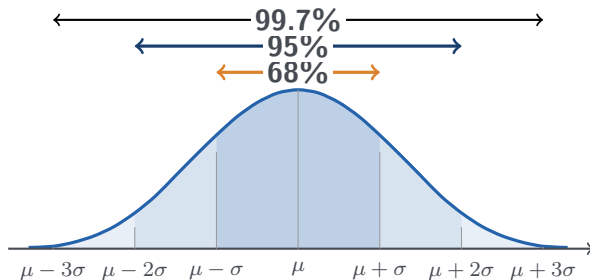
Interval	Approximate percentage
$\mu \pm 1\sigma$	68%
$\mu \pm 2\sigma$	95%
$\mu \pm 3\sigma$	99.7%

Condition

Use the empirical rule only when the distribution is approximately bell-shaped.

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.

Section 3.4.2: The 68–95–99.7 Rule



Empirical Rule for a bell-shaped distribution

$$\mu \pm \sigma : 68\%$$

$$\mu \pm 2\sigma : 95\%$$

$$\mu \pm 3\sigma : 99.7\%$$

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.

Example 3-22: Age Distribution of Persons

Problem

The age distribution of a sample of 5000 persons is bell-shaped with a mean of 40 years and a standard deviation of 12 years. Determine the approximate percentage of people who are between 16 and 64 years old.

$$\bar{x} = 40, \quad s = 12.$$

Both endpoints are 24 years from the mean:

$$40 - 16 = 24, \quad 64 - 40 = 24.$$

Convert this distance to standard deviations:

$$\frac{24}{12} = 2.$$

Thus the interval is

$$\bar{x} - 2s \text{ to } \bar{x} + 2s.$$

Empirical Rule

Key Takeaways

- Range = largest value minus smallest value.
- Variance and standard deviation measure spread around the mean.
- Sample data use s ; population data use σ .
- On the TI-84, S_x is the sample SD and σ_x is the population SD.
- Standard deviation has the same units as the original data.
- For bell-shaped data: 68%, 95%, and 99.7% lie within 1, 2, and 3 standard deviations of the mean.

Course note

Coefficient of variation (Section 3.2.3) and grouped-data calculations are intentionally omitted.

Based on: Prem S. Mann, *Introductory Statistics*, 10th ed., Wiley, Sections 3.2 and 3.4.2.